What is cooking?
What is XNLP?
What is XNLP-complete?
Counter Check Machines
Shuffle Product
Directed Bandwidth
Discussion
References

Solo Cooking is XNLP-complete

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October 7, 2026

Plan

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2. What is XNLP?
3. What is XNLP-complete?
   - Counter Check Machines
   - Shuffle Product
   - Directed Bandwidth
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Cooking time!

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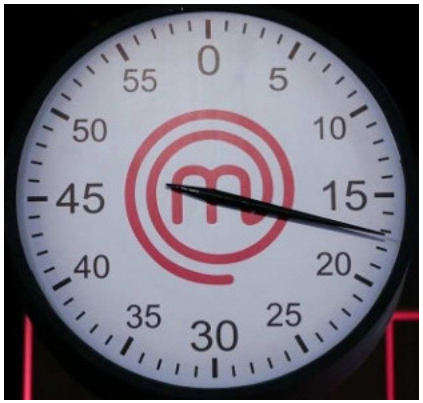


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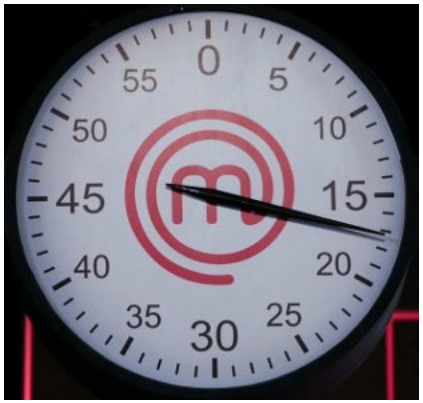
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Cooking = set of tasks ruled by precedence and waiting time constraints.

- Between tasks: a range $[a, b]$ of available waiting time values.

Cooking a definition

Solo Cooking

Given: a set of n elementary tasks $\mathcal{T} = \{T_1, \ldots, T_n\}$ of length $p_1, \ldots, p_n$,
a directed acyclic graph $G = (\mathcal{T}, A)$,
two nonnegative integers D and ℓ,
for each arc $(i, j) \in A$:

- a min. waiting time constraint $\ell_{i,j}^{min} \in [0, \ell]$,
- a max. waiting time constraint $\ell_{i,j}^{max} \in [0, \ell] \cup \{\infty\}$.

Question: is there a schedule which completes all tasks in $\mathcal{T}$ by time D one at a time while meeting all constraints?

- (Single machine scheduling with precedence delays)
- NP-complete [Garey et al., 1978, Leung et al., 1984].

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Cooking parameters

Maximum task length p_{max}.

Waiting time range parameter ℓ.

Width w.

Why solo cooking?

Theorem [van Bevern et al., 2016]

With two cooks and no waiting time restriction (i.e., $\ell = 0$ and range $[0, \infty)$ for all arcs), the problem is:

- NP-complete with three chains of arbitrary-length tasks,
- W[2]-hard parameterized by w with tasks of length 1 or 2.

Theorem [Mallem et al., 2026, Bodlaender and Mallem, 2026]

Solo Cooking (and several of its restrictions) are XNLP-complete parameterized by $\ell + w$.

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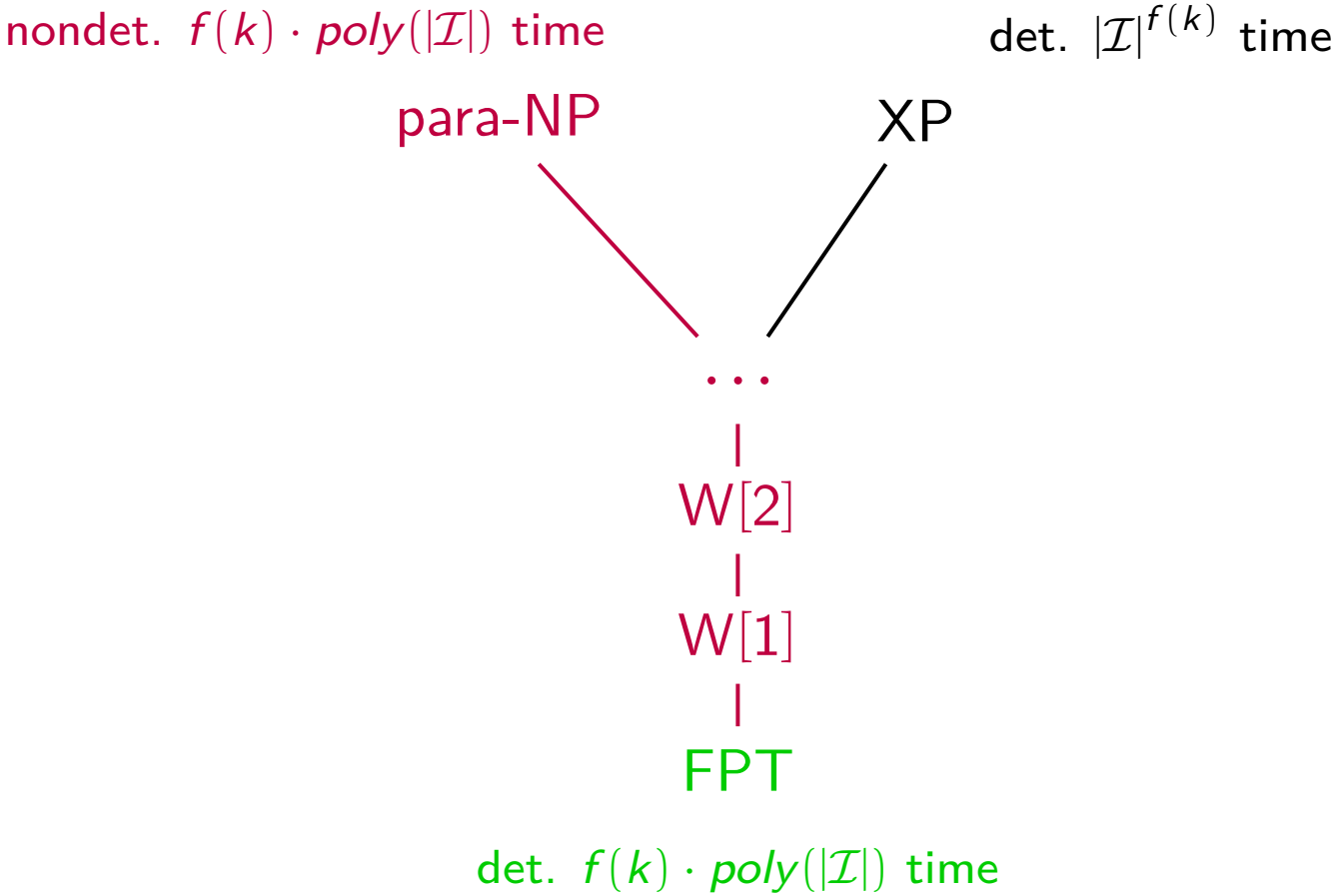
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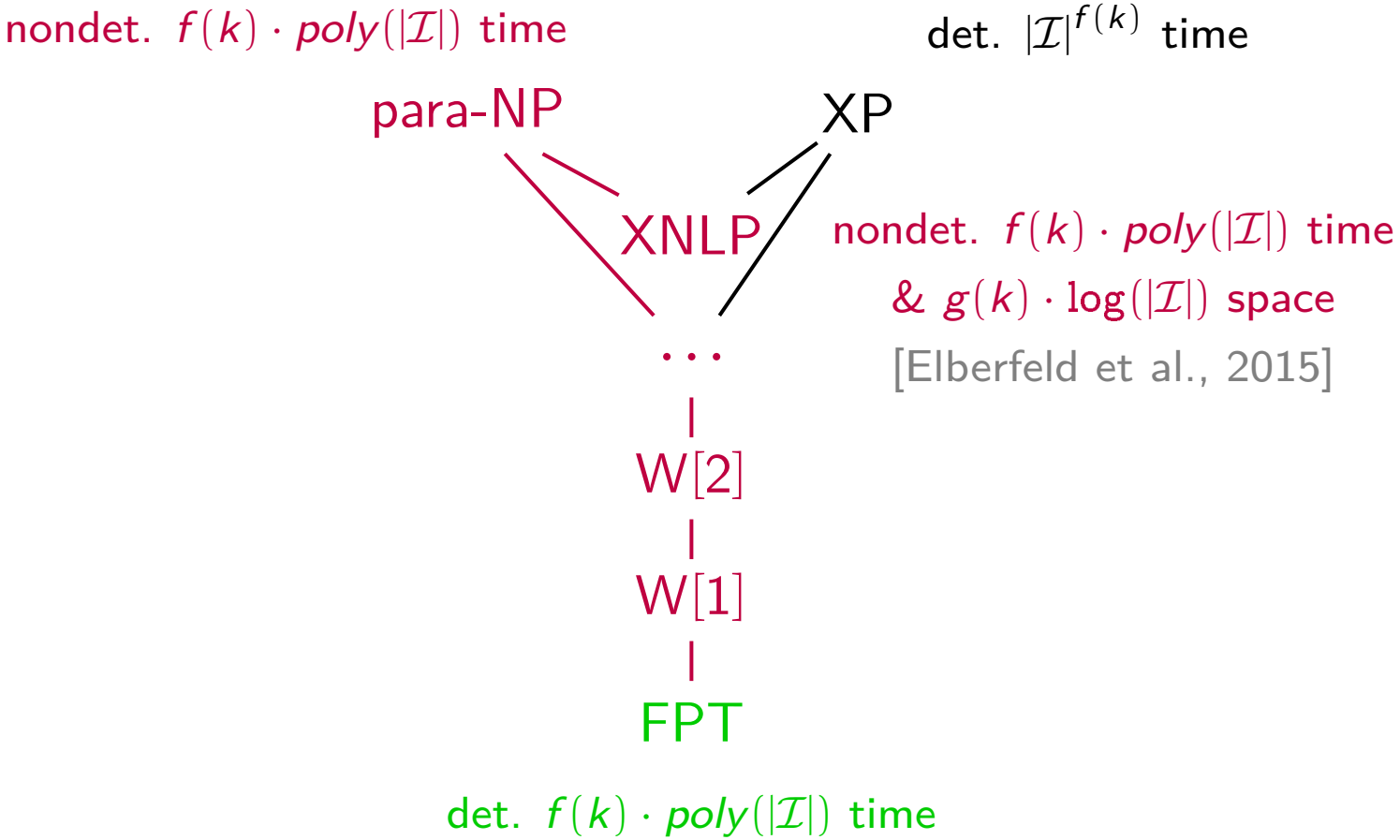
Parameterized complexity classes

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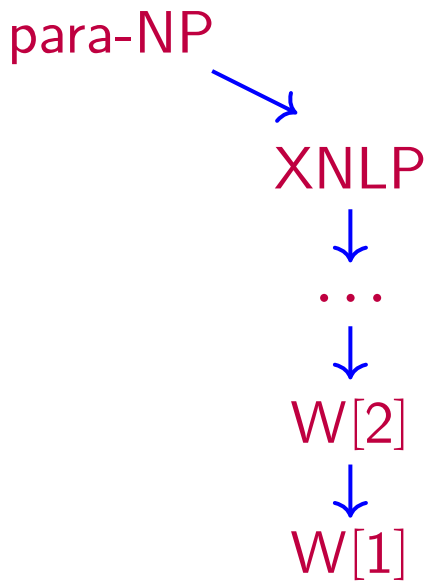
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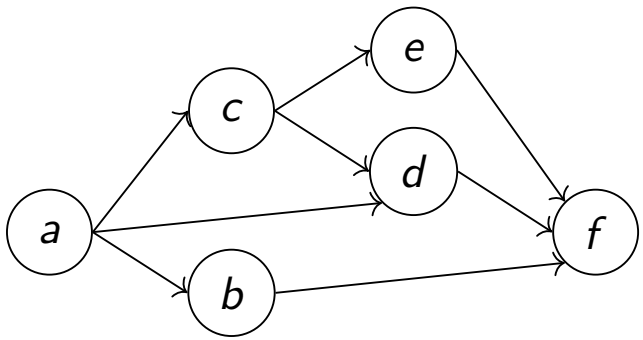
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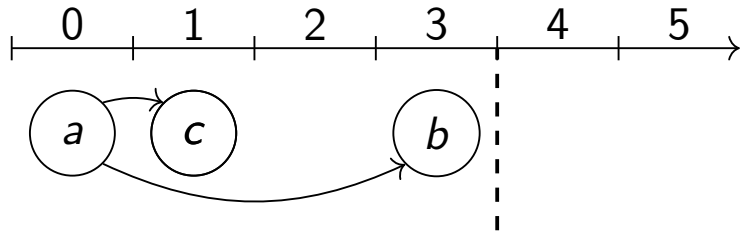


nondet. $f(k) \cdot poly(|\mathcal{I}|)$ time
& $g(k) \cdot \log(|\mathcal{I}|)$ space
[Elberfeld et al., 2015]

Solo Cooking is in XNLP

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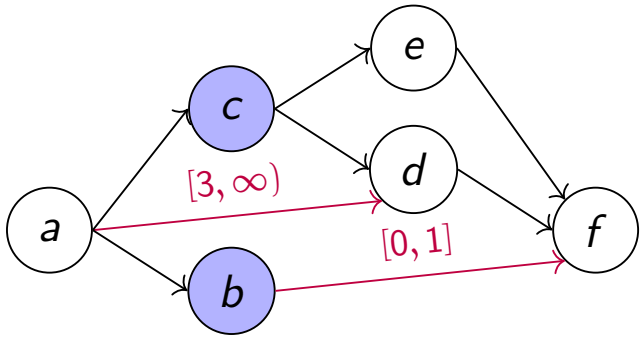


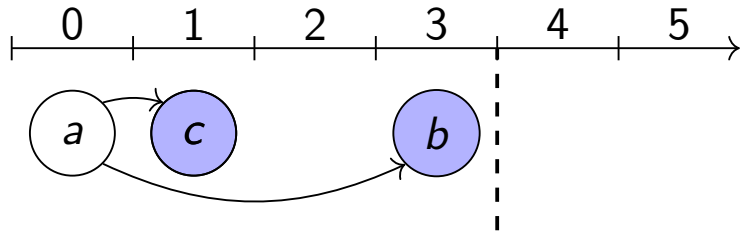


State

- Current time: $\mathcal{O}(\log(n\ell p_{max}))$ space.
  - Here: 4.
- Border of completed tasks: $\mathcal{O}(w \log(n))$ space.
  - Here: $\{b, c\}$.
- Schedule of completed tasks with pending waiting time constraints: $\mathcal{O}(\ell \log(n\ell p_{max}))$ space.
  - E.g., $\{(a, 0), (b, 3)\}$ if (c, d) and (c, e) have no waiting time constraints.

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Nondeterministic Nondecreasing Counter Check Machine (NNCCM)

NNCCM Acceptance

Given: the description of a NNCCM:

- number of counters k, max. counter value n, number of checks r,
- a sequence of r checks in $([1,k] \times [0,n] \times [1,k] \times [0,n])^r$.

Parameter: number of counters k.

Question: is there an accepting run of the NNCCM?

Example: $k = 3$, $n = 2$, $r = 2$ checks: $((1,0,2,0), (1,1,3,0))$.

Example run 1:

	r_1	r_2
c_1	0	1	1
c_2	0	0	0
c_3	0	0	1

Example run 2:

	r_1	r_2
c_1	0	2	2
c_2	0	0	0
c_3	0	0	0

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(Binary) Shuffle Product

Given: source words $s_1, \ldots, s_k$ and target word t over a (binary) alphabet Σ.

Parameter: number of source words k.

Question: is t in the shuffle product of $s_1, \cdots, s_k$?

$$
\begin{array}{rccccccccc}
s_1 = & c & b & a & a & & & & & \\
s_2 = & a & b & c & & & & & & \\
s_3 = & c & a & & & & & & & \\
t = & a & c & b & b & c & a & a & a & c
\end{array}
$$

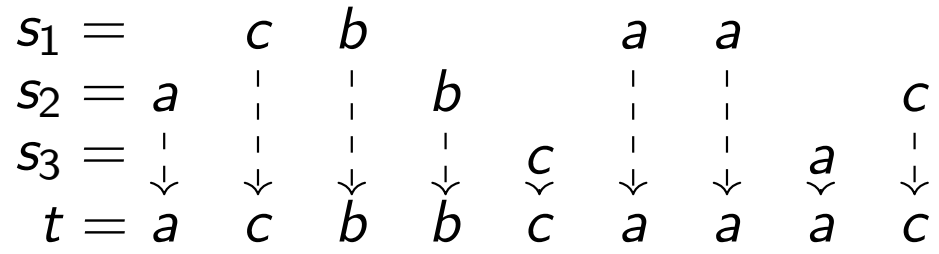
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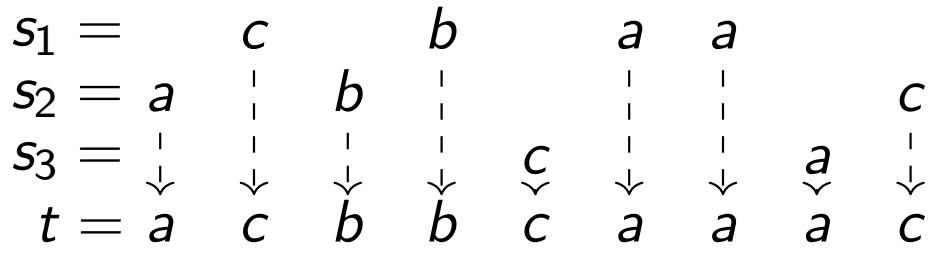
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Reduction for (Binary) Shuffle Product

Example: $k = 3$, $n = 2$, $r = 2$ checks: $((1, 0, 2, 0), (1, 1, 3, 0))$.

	r_1	r_2
c_1	0	0
c_2	0	0
c_3	0	0

	r_1	r_2
c_1	0	1
c_2	0	0
c_3	0	0

	r_1	r_2
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$$
\begin{aligned}
s_1 &= a\ b\ a\ b\ a\ bb\ a\ b \quad (\dots) \quad a\ b\ a\ bb\ a\ b\ a\ b \quad (\dots) \quad b \\
s_2 &= a\ b\ a\ b\ a\ bb\ a\ b \quad a\ b\ a\ b\ a\ b\ a\ b \quad b \\
s_3 &= a\ b\ a\ b\ a\ b\ a\ b \quad a\ b\ a\ b\ a\ bb\ a\ b \quad bb \\
t &= a^3\, b^5\, a^3\, b^5\, a^3\, b^4\, a^3\, b^5 \quad a^3\, b^5\, a^3\, b^5\, a^3\, b^4\, a^3\, b^5 \quad b^4\, a^3\, b^5\, a^3\, b^5
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$s_1 = a\ b \quad a\ b\ a\ bb$ $\quad a\ b\ a\ b\ a\ bb\ a\ b \quad bb\ a\ b$

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\quad (\dots)
$$

$$
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\end{aligned}
\quad (\dots)
$$

$$
\begin{aligned}
&bb\,a\ b \\
&b \\
&b\ a\ bb \\
&b^4\, a^3\, b^5\, a^3\, b^5
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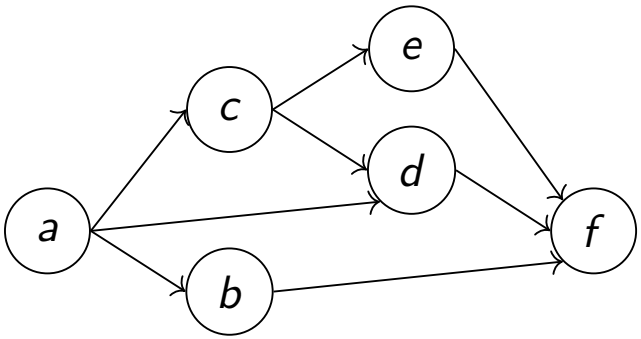
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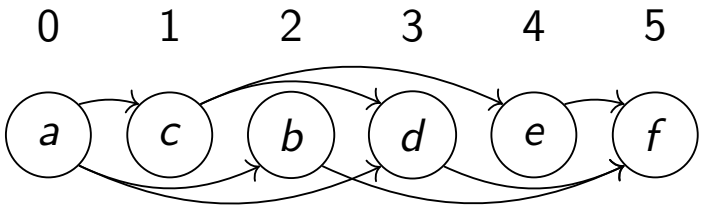
Directed Bandwidth

Given: a directed acyclic graph $G = (V, A)$.

Parameter: target value B.

Question: is there $f : V \to \{0, \cdots, |V| - 1\}$ a bijective mapping such that, for each arc (i, j) in A, $f(j) - f(i) \leq B$?





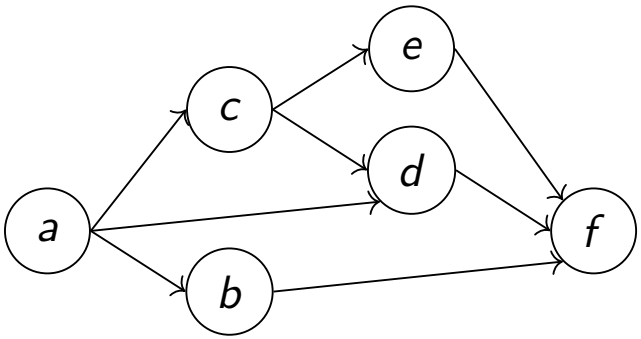
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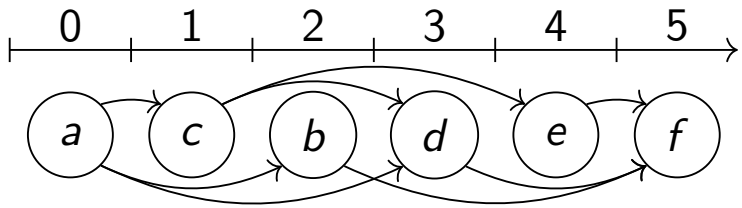
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- Special case of Solo Cooking!
  - Unit tasks, deadline $D = n$ and waiting time in $[0, B - 1]$ for all (i, j) in A.

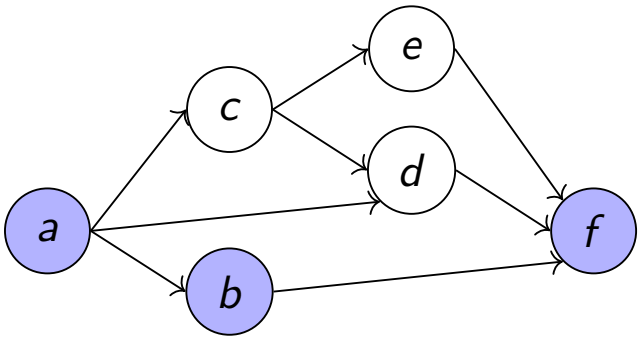
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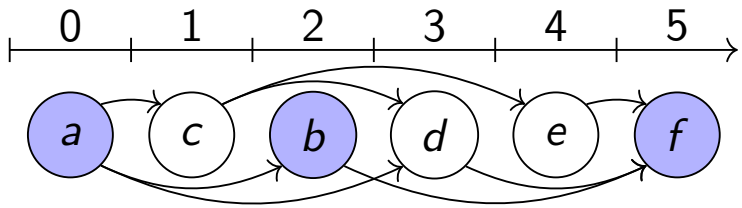
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Takeaway

- Even alone, cooking is hard.
- Your favorite problem might be XNLP-complete too!

Going further

- Accepting NNCCM, List Coloring, Chained Multicolored Clique, etc. [Bodlaender et al., 2024].
- Graphs with a linear structure [Bodlaender et al., 2025].
- Planar graphs [Bodlaender and Szilágyi, 2026].
- Dominating / Independent set reconfiguration [Bodlaender et al., 2026].

Open questions

- Solo cooking of unit tasks with equal-length waiting times.
  - $\leftrightarrow$ Unary Bin Packing with a bounded number of bins.
  - W[1]-hard [Jansen et al., 2013], in XP [Garey and Johnson, 1979].
- $[\ell_{i,j}^{min}, \ell_{i,j}^{max}) = [\ell, \infty)$ for all (i, j) in A.
  - $\leftrightarrow Pm|prec, p_j = 1|C_{max}$ (open since [Ullman, 1975]).
  - XNLP-hard [Bodlaender et al., 2022].

Discussion

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  - $\leftrightarrow$ Unary Bin Packing with a bounded number of bins.
  - W[1]-hard [Jansen et al., 2013], in XP [Garey and Johnson, 1979].
- $[\ell_{i,j}^{min}, \ell_{i,j}^{max}) = [\ell, \infty)$ for all (i, j) in A.
  - $\leftrightarrow Pm|prec, p_j = 1|C_{max}$ (open since [Ullman, 1975]).
  - XNLP-hard [Bodlaender et al., 2022].

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